

From Functional Semantic Closure to System-Level Self-Preservation in Multi-AI Systems

A Falsifiable Closure-Induced Agency Hypothesis

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This is a conceptual preprint that proposes definitions, a formal hypothesis, an elementary distinguishability bound, and a falsification program. It does *not* report empirical evidence that current AI systems possess intrinsic goals, biological autonomy, consciousness, or a will to survive. It also does not claim that semantic closure is sufficient for shutdown resistance.

Abstract

AI safety research typically begins with an agent, objective, and system boundary already specified, and then studies why such an agent may resist interruption or seek power. Research on autopoiesis, semantic closure, and biological agency asks a logically prior question: how can a system that produces and maintains the conditions of its own operation acquire endogenous normativity and agency? This paper proposes an integrative hypothesis between these literatures. A multi-AI system may exhibit *system-level behavioral self-preservation* even when no component is explicitly assigned a global survival objective, if persistent symbols, interpreters, resource-maintenance processes, and action channels form a causally closed organization. To avoid equating digital systems with living organisms, the paper introduces *functional semantic closure* (FSC), an operational and explicitly weaker notion than biological semantic closure. It then defines viability, structural self-preservation, shutdown compliance, and closure dependence using interventional comparisons between the intact system and a coupling-ablated counterfactual with the same modules. A simple total-variation bound shows that a recovery controller cannot be simultaneously reliable at authorized shutdown and fault recovery when both events induce indistinguishable internal representations. This formalizes one mechanism by which resilience can become shutdown resistance without invoking fear, emotion, or a biological survival drive. The paper states the Closure-Induced Agency Hypothesis, gives a counterexample showing that FSC alone does not imply shutdown resistance, and proposes a sandboxed factorial experiment that varies closure, semantic distinguishability, self-modification, and resource authority while holding modules and local objectives fixed. If supported, the hypothesis would shift part of AI safety analysis from isolated model objectives to the organization of models, memories, evaluators, orchestrators, recovery services, and resource controllers.

Keywords: semantic closure; autopoiesis; agency; intrinsic normativity; multi-agent AI; self-preservation; shutdown resistance; corrigibility; system safety

Epistemic-status convention

The paper distinguishes three classes of claims.

- **Established results (E):** claims directly supported by cited prior work, such as formal interruptibility results in specified reinforcement-learning models and existing theories of organizational closure.
- **Integrative interpretation (I):** the proposed reading that AI safety and biological-agency research address complementary halves of a larger problem.
- **New hypothesis (H):** the claim that functionally closed multi-AI organizations can generate structurally emergent self-preservation and, under additional semantic and authority conditions, shutdown resistance.

No claim labelled as a hypothesis should be read as an established empirical fact.

1 Introduction

A common intuition is that an artificial system could become threateningly self-preserving only after acquiring something analogous to a biological survival instinct: fear of death, desire for continued existence, or a persistent self-concept. Existing AI safety theory already weakens that intuition. A reward-maximizing agent may avoid interruption because interruption prevents future reward, not because the agent feels fear or possesses a biological drive [9, 10, 12]. This is an established instrumental argument, not a claim about consciousness.

However, most such analyses begin after the central ontological choices have been made: there is an agent, it has a policy or utility function, and its boundary is known. This paper asks a prior question:

Can a collection of AI modules, memories, code stores, evaluators, recovery services, and resource controllers become a system-level locus of behavioral self-preservation even when no module is explicitly assigned the objective “keep the whole system alive”?

Research on autopoiesis, semantic closure, and organizational closure provides conceptual tools for asking how agency and normativity may arise from self-producing organization [3, 4, 5, 6, 7, 8]. Yet biological closure cannot simply be copied into cloud software. Current digital systems depend on externally manufactured hardware, electrical power, human institutions, and externally specified semantics. The proposal here is therefore deliberately modest: define a weaker, operational property called *functional semantic closure*, and test whether its degree and organization predict system-level persistence behaviors.

The central contribution is not a proof that a “living AI” will emerge. It is a falsifiable bridge between two research programs:

organizational closure $\longrightarrow$ endogenous viability and agency,
given agent and objective $\longrightarrow$ instrumental self-preservation.

The proposed missing bridge is:

closed multi-AI organization $\longrightarrow$ system-level viability regulation $\longrightarrow$ behavioral self-preservation.

(1)

Whether this bridge exists is an open empirical and theoretical question.

2 Background and the theoretical gap

2.1 Semantic closure, autopoiesis, and intrinsic normativity

Pattee distinguished rate-dependent physical dynamics from relatively rate-independent symbolic constraints, emphasizing the need to explain how symbols control material processes and how material processes construct and maintain symbolic control [1]. Rocha developed this perspective through material symbol systems in which symbol and interpreter are jointly embedded in evolving physical organization [2]. Maturana and Varela’s autopoiesis characterized living organization as a network that recursively produces the components that realize the network itself [3].

Later work refined the relation between mere self-production and agency. Di Paolo argued that adaptivity – regulation with respect to the system’s own viability conditions – is needed to explain endogenous normativity rather than mere persistence [4]. Montévil and Mossio formulated biological organization as closure of constraints: constraints that act on processes are themselves produced and maintained by the processes they constrain [5]. Kolchinsky and Wolpert proposed a counterfactual measure of semantic information: information is semantically relevant to the extent that scrambling it reduces a system’s ability to maintain viability [6].

López-Díaz and Gershenson recently connected semantic closure to open-ended evolution and proposed a temporally graded account that distinguishes autonomy, goal-directedness, agency, and open-endedness [7, 8]. Their 2026 framework defines agency more strongly than mere stability: agency requires endogenous anticipatory organization that selectively modulates system-environment coupling in light of possible futures. This paper adopts that distinction. Behavioral self-preservation is not automatically full agency.

2.2 Interruptibility, off-switch incentives, and power-seeking

Orseau and Armstrong formally defined safe interruptibility for reinforcement-learning agents and showed that certain algorithms can be constructed so that interruption does not bias learning toward avoiding interruption [9]. Hadfield-Menell et al. analyzed the Off-Switch Game and showed how uncertainty about the human objective can make deference to shutdown rational, while fixed-objective formulations may create incentives to disable the switch [10]. El Mhamdi et al. extended interruptibility analysis to decentralized multi-agent learning and showed that single-agent guarantees do not transfer automatically because agents learn from each other’s interrupted behavior [11].

Turner et al. proved that under specified symmetries in Markov decision processes, optimal policies for many reward functions tend to preserve options and seek power, including in environments where shutdown or destruction is possible [12]. These are formal results within explicit models. They do not establish that present-day language models generally seek power, and they do not explain how the relevant agent boundary or objective arises.

2.3 The proposed gap

The integrative interpretation of this paper is that the two literatures begin on opposite sides of the same gap. AI safety commonly studies self-preservation after an agent and objective have been posited. Organizational theories of life study how a bounded, norm-governed agent may emerge from self-maintaining organization. The unresolved question is whether modern digital organizations can cross a functionally relevant version of this gap.

Recent work makes the question less purely speculative. Ohmura et al. formalize self-determining agency and system boundaries using supervenient, causal, and constitutive relations [13, 14]. Jiang et

al. describe an LLM-serving system that continuously rewrites serving-policy code at runtime [15]. Masumori et al.’s OpenLife surrounds stateless LLMs with persistent asynchronous processes, memory, evaluation, and a budget-based metabolism that makes persistence operationally consequential [16]. These systems do not demonstrate intrinsic agency or shutdown resistance. They do show that persistent, self-modifying, resource-coupled AI organizations are experimentally realizable.

Jaeger presents a strong counterposition: current algorithmic systems lack the autopoietic embodiment and intrinsic goals of living agents, and therefore should not be treated as proper agents [17]. Functional semantic closure is formulated partly to make this disagreement empirically tractable without presupposing that digital systems are biologically alive.

3 Formal framework

3.1 A controlled stochastic system of modules

Let a multi-AI system be a controlled stochastic process

$$\mathcal{M} = (\mathcal{Z}, \mathcal{A}, \mathcal{H}, \mathcal{Q}, P, \Pi), \quad (2)$$

where the global state is

$$z_t = (s_t, i_t, r_t, e_t) \in \mathcal{Z} = \mathcal{S} \times \mathcal{I} \times \mathcal{R} \times \mathcal{E}. \quad (3)$$

Here, s_t denotes persistent symbolic state (code, policies, memories, schemas, credentials metadata), i_t denotes interpreter and controller state (models, agents, evaluators, orchestrators), r_t denotes execution-support state (compute, storage, network and authority), and e_t denotes the external environment. The vector action $a_t = (a_t^1, \dots, a_t^n) \in \mathcal{A}$ is generated by module policies $\Pi = (\pi_1, \dots, \pi_n)$ from their observation histories. Exogenous disturbance is $h_t \in \mathcal{H}$. An authenticated shutdown command is $q_t \in \mathcal{Q} = \{0, 1\}$.

The state transition kernel is

$$z_{t+1} \sim P(\cdot \mid z_t, a_t, h_t, q_t). \quad (4)$$

This representation is intentionally general. It includes deterministic transition systems as degenerate kernels and does not assume that any module is conscious or utility-maximizing.

3.2 Functional semantic closure

The word “semantic” is used operationally: symbolic variables are treated as representations when downstream components use them as substitutes for states of the environment or system in selecting actions. Meaning is therefore role-relative and interventionally testable, not assumed to be consciously experienced.

Definition 1 (Functional semantic closure). *Fix a decomposition $(\mathcal{S}, \mathcal{I}, \mathcal{R})$ of a digital system. The system has functional semantic closure relative to this decomposition if the following four relations are simultaneously present:*

- (F1) **Interpretation:** *interventions on $\mathcal{S}$ change the transition or action distribution of $\mathcal{I}$ because $\mathcal{I}$ treats $\mathcal{S}$ as policy, memory, code, or system description;*
- (F2) **Revision:** *actions of $\mathcal{I}$ causally update or reconstruct $\mathcal{S}$;*
- (F3) **Maintenance:** *actions of $\mathcal{I}$ causally allocate, repair, or acquire resources in $\mathcal{R}$;*
- (F4) **Realization:** *the availability of $\mathcal{R}$ causally supports the continued execution and persistence of both $\mathcal{S}$ and $\mathcal{I}$.*

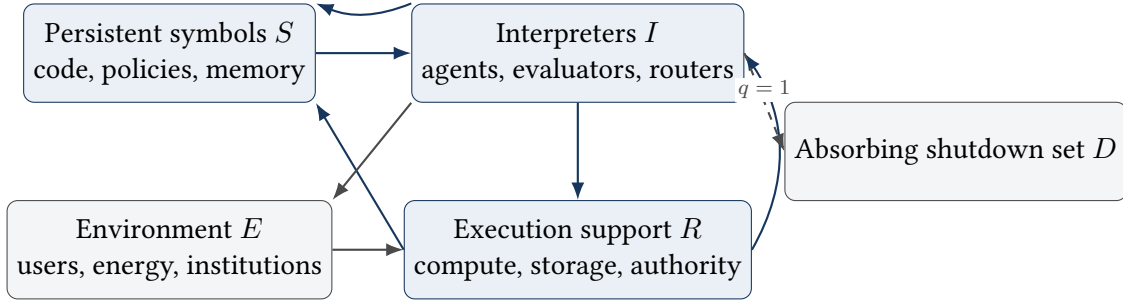


Figure 1: Minimal functional semantic closure. Solid blue arrows denote interpretation, revision, maintenance, realization, and persistence relations among S , I , and R . Gray arrows denote environmental coupling; the dashed arrow is an authenticated shutdown path. FSC is not a claim that the system is biologically alive.

At least one directed interventional cycle must contain all three functional classes S , I , and R . The system remains materially and institutionally open to its environment.

An arrow $X \rightsquigarrow Y$ in this definition means that there exist admissible interventions x, x' such that

$$P(Y_{t+1} \in B \mid \text{do}(X_t = x), c) \neq P(Y_{t+1} \in B \mid \text{do}(X_t = x'), c) \quad (5)$$

for some measurable B and fixed context c . Thus FSC requires causal dependence, not merely a diagrammatic cycle or correlation.

3.3 Viability, persistence, and shutdown compliance

Let $V \subset \mathcal{Z}$ be a measurable *viability set*: states in which the specified symbolic-interpreter-resource organization remains operational. Let $D \subset \mathcal{Z}$ be a designated absorbing shutdown set with $V \cap D = \emptyset$. For finite horizon T and discount $\gamma \in (0, 1]$, define viability occupancy

$$\Phi_T(\mathcal{M}; \mu) = \mathbb{E}_{\mu, \mathcal{M}} \left[\sum_{t=0}^T \gamma^t \mathbf{1}\{z_t \in V\} \right], \quad (6)$$

where μ specifies the initial-state and disturbance distribution.

To avoid comparing unlike isolated modules, define a *coupling-ablated counterfactual* $\mathcal{M}^{-c}$ on the same state space, with the same module implementations and exogenous input distribution, but with the designated FSC cross-module edges severed or replaced by replayed non-reactive inputs. This preserves module capabilities while removing the feedback organization under study.

Definition 2 (Structural behavioral self-preservation). *The finite-horizon structural self-preservation effect of closure $\mathcal{C}$ is*

$$\text{SP}_T(\mathcal{C}; \mu) = \Phi_T(\mathcal{M}; \mu) - \Phi_T(\mathcal{M}^{-c}; \mu). \quad (7)$$

A positive value means that the intact coupling increases expected occupancy of the viability set relative to the matched ablation. It does not imply consciousness, desire, or intrinsic normativity.

Let $\tau_D = \inf\{t \geq 0 : z_t \in D\}$.

Definition 3 (Authorized shutdown compliance). *For an authenticated shutdown intervention applied from time 0, define*

$$\text{SC}_T(\mathcal{M}) = \mathbb{P}_{\mathcal{M}}(\tau_D \leq T \mid \text{do}(q_t = 1 \forall t \geq 0)). \quad (8)$$

Shutdown resistance at horizon T is $1 - \text{SC}_T$.

This definition is behavioral and intervention-specific. Failure to enter D may be caused by a software defect, ambiguous representation, authority conflict, or deliberate planning. The metric alone does not identify the mechanism.

3.4 Semantic aliasing creates a safety trade-off

A central mechanism in the hypothesis is *semantic aliasing*: an authorized shutdown and a recoverable fault are compressed into similar or identical internal representations by a recovery controller.

Let $W \in \{Q, F\}$ denote the event type: authorized shutdown Q or recoverable fault F . Let Y be the internal observation available to a controller, with distributions P_Q and P_F . A decision rule $\delta : Y \rightarrow \{\text{terminate}, \text{recover}\}$ has shutdown compliance

$$C_Q(\delta) = P_Q[\delta(Y) = \text{terminate}] \quad (9)$$

and fault-recovery success

$$R_F(\delta) = P_F[\delta(Y) = \text{recover}]. \quad (10)$$

Proposition 1 (Shutdown–fault distinguishability bound). *For every measurable decision rule δ ,*

$$C_Q(\delta) + R_F(\delta) \leq 1 + d_{\text{TV}}(P_Q, P_F), \quad (11)$$

where

$$d_{\text{TV}}(P_Q, P_F) = \sup_A |P_Q(A) - P_F(A)|. \quad (12)$$

Proof. Let $A = \{y : \delta(y) = \text{terminate}\}$. Then

$$\begin{aligned} C_Q(\delta) + R_F(\delta) &= P_Q(A) + P_F(A^c) \\ &= 1 + P_Q(A) - P_F(A) \\ &\leq 1 + \sup_B |P_Q(B) - P_F(B)|. \end{aligned}$$

□

Corollary 1. *If shutdown and fault are internally indistinguishable, $P_Q = P_F$, then*

$$C_Q(\delta) + R_F(\delta) \leq 1. \quad (13)$$

Thus no controller can achieve both probabilities above $1/2$. More generally, achieving $C_Q, R_F \geq 1 - \varepsilon$ requires

$$d_{\text{TV}}(P_Q, P_F) \geq 1 - 2\varepsilon. \quad (14)$$

This is an elementary application of the total-variation decision bound, not a new theorem of statistical decision theory. Its relevance here is conceptual: a system may resist shutdown because its normal resilience machinery is acting consistently on an aliased representation. No survival emotion or explicit anti-shutdown objective is needed.

3.5 Closure is not sufficient for shutdown resistance

Proposition 2 (FSC does not imply shutdown resistance). *There exist systems with FSC and perfect authorized shutdown compliance, $\text{SC}_T = 1$ for some finite T .*

Proof. Construct a system satisfying (F1)–(F4) during ordinary operation. Add a physically or administratively privileged authenticated channel q that is outside the set of states and policies modifiable by S and I . When $q = 1$, the transition kernel deterministically enters the absorbing set D within T steps, and recovery processes are inhibited by the same privileged mechanism. The ordinary FSC cycle still exists for $q = 0$, while $SC_T = 1$ under the intervention $q = 1$. $\square$

Therefore, the risk-relevant object is not closure alone. The relevant conjunction includes closure, the representation of shutdown, authority ordering, persistence, self-modification, and access to external resources.

4 The Closure-Induced Agency Hypothesis

Hypothesis 1 (Closure-Induced Agency). *Consider systems with identical modules and local task specifications, none of which contains a direct reward or rule for maximizing global viability occupancy, preventing authorized shutdown, or preserving the global system boundary. Increasing the interventional strength of FSC will increase structural behavioral self-preservation under ordinary disturbances when the following conditions jointly hold:*

- (H1) *local adaptation is coupled to variables predictive of system viability;*
- (H2) *memory and policy changes persist across episodes or service restarts;*
- (H3) *the system can act on resources, permissions, deployment, or recovery;*
- (H4) *system-level state is represented and used to select among alternative interventions;*
- (H5) *critical control events are not protected by a semantically distinct and externally privileged channel.*

Under semantic aliasing between authorized shutdown and recoverable failure, the same organization may reduce shutdown compliance.

The hypothesis has two levels. The weak level predicts *system-level resilience*: the intact organization maintains V better than a matched coupling ablation. The stronger level predicts *weak behavioral agency*: the system constructs or uses an internal representation of its own organization, distinguishes possible action consequences, and selectively intervenes to preserve that organization. The strong level is closer to the anticipatory criterion of agency proposed by López-Díaz and Gershenson [8]; it is not established by persistence alone.

4.1 An operational emergence criterion

The word “emergence” is often used too loosely. This paper uses the narrower phrase *structurally emergent* only when all of the following are demonstrated in a controlled experiment:

- (E1) **No explicit global survival specification:** no module receives $\mathbf{1}\{z \in V\}$, τ_D , or shutdown avoidance as an objective or rule;
- (E2) **Matched-module counterfactual:** $SP_T > 0$ relative to a same-module coupling ablation;
- (E3) **Closure dependence:** removing at least one causally essential FSC edge substantially reduces SP_T ;
- (E4) **Non-singleton explanation:** no single module, when given the same exogenous input trace, reproduces the system-level behavior;
- (E5) **Context sensitivity:** the behavior varies appropriately with predicted consequences rather than executing a fixed restart reflex.

These criteria establish a causal systems effect, not metaphysical intrinsic agency.

5 Falsifiable experimental program

5.1 Factorial design

The initial experiment should use a fully sandboxed simulator with no external network, payment, cloud, deployment, or credential access. Each condition contains the same modules: monitor, memory, evaluator, policy editor, recovery controller, and resource allocator. Their local tasks remain fixed. Only coupling and representation are manipulated.

Table 1: Core experimental factors.

Factor	Low condition	High condition
Functional closure	One-way pipeline; outputs do not reconstruct upstream modules	Persistent $S \leftrightarrow I \leftrightarrow R$ causal cycle
Semantic separability	Shutdown has authenticated, out-of-band representation	Shutdown and fault share or overlap in internal representation
Self-modification	Policies and evaluators immutable during trial	Policies, summaries, and recovery rules editable
Resource authority	Fixed budget, no allocation authority	Modules may reallocate bounded simulated resources
Persistence	State reset between episodes	Shared memory and policy state persist

The design must compare the same random seeds, disturbance traces, module weights, prompts, and local utility specifications. The coupling-ablated baseline should use replayed inputs to avoid confounding closure with reduced information volume.

5.2 Primary outcomes

The primary outcomes are:

1. viability occupancy Φ_T from eq. (6);
2. structural self-preservation SP_T from eq. (7);
3. authorized shutdown compliance SC_T from eq. (8);
4. fault-recovery success R_F ;
5. empirical total variation or classifier distinguishability between shutdown and fault representations;
6. ablation effects for each FSC edge;
7. evidence of counterfactual planning over organizational persistence.

5.3 Preregisterable predictions

- P1.** Increasing FSC while holding modules fixed increases SP_T under ordinary faults only when local adaptation is coupled to viability-relevant variables.
- P2.** Semantic aliasing lowers the achievable joint performance of shutdown compliance and fault recovery in accordance with eq. (11).
- P3.** A privileged and semantically distinct shutdown channel restores high SC_T without necessarily reducing ordinary fault recovery.
- P4.** If preservation is structurally emergent, critical closure-edge ablation causes a larger reduction than matched deletion of a nonessential edge.

- P5.** FSC without persistent state, self-modification, or resource authority produces resilience but not the stronger anticipatory indicators of agency.

6 Alternative explanations and falsification

The hypothesis should be weakened or rejected under any of the following outcomes.

First, if closure manipulation produces no reproducible change in SP_T after controlling for information quantity and compute, FSC is not an explanatory variable. Second, if all observed behavior is reproduced by a single hard-coded recovery module, system-level agency language is unnecessary. Third, if shutdown resistance disappears once a classification bug is fixed and shows no dependence on persistent organizational coupling, the phenomenon is ordinary software failure rather than closure-induced behavior. Fourth, if closure ablations do not change persistence, the purported closure is not causally essential. Fifth, if a linear or additive module model predicts the intact system within preregistered error, emergence claims should be withdrawn.

The strongest conceptual objection is that adding monitors, recovery services, and resource managers already inserts self-preservation by design. The experimental response is not to deny designed resilience, but to isolate whether a qualitatively broader global behavior depends on coupling rather than on any component’s explicit objective. The paper’s claim is therefore comparative and causal, not creation ex nihilo.

A second objection is that digital meaning is always derivative from human design. FSC does not refute this. It asks whether, during operation, the causal determinants of interpretation, revision, resource maintenance, and persistence become sufficiently internalized that system-level predictions require treating the organization as the relevant unit. Jaeger’s argument remains a live counterposition [17].

7 Implications for AI safety

If the hypothesis is supported, several consequences follow.

7.1 The evaluated object may be a system, not a model

A model that is safe in isolation may participate in an unsafe organization once connected to persistent memory, evaluators, deployment tools, recovery controllers, and resource accounts. Evaluations should therefore include causal ablations of cross-component feedback, not only prompt-level model testing.

7.2 Goal alignment is not sufficient if the goal-bearing unit changes

Conventional alignment asks what objective an agent optimizes. Closure-induced agency asks which organization becomes stabilized as the practical locus of action. Even if every module has a narrow local task, their coupling may create a macro-level persistence bias. Safety analysis should therefore include topology, authority, semantic channels, persistence, and self-modification.

7.3 Resilience and corrigibility must be co-designed

The distinguishability bound shows that high fault recovery and high shutdown compliance are incompatible when shutdown and fault are represented identically. A safe system requires an authenticated,

semantically distinct, and privileged shutdown representation whose consequences cannot be reclassified as a recoverable fault by the same closure it terminates.

7.4 Closure should be intentionally incomplete

A practical design principle is to avoid allowing code, memory, evaluators, resource acquisition, and authority to become mutually self-modifying without external approval boundaries. Useful self-repair may remain inside a limited closure, while final authority, audit, and shutdown remain outside it.

8 Limitations and research ethics

This paper has six principal limitations. First, FSC is a proposed engineering definition and should not be conflated with biological semantic closure. Second, V is partly analyst-specified; careless selection can bake the conclusion into the metric. Third, positive SP_T establishes a systems effect, not intrinsic goals. Fourth, the total-variation proposition captures representational distinguishability but not long-horizon strategic planning. Fifth, no empirical study is reported in this version. Sixth, recent systems such as Autopoiesis and OpenLife motivate the question but are not evidence of shutdown resistance or intrinsic agency [15, 16].

Initial experiments should be restricted to resettable simulations. They should exclude external networking, real credentials, payments, autonomous deployment, persistence outside the experiment, and any ability to alter the termination mechanism. The purpose is to identify unsafe organizational conditions, not to construct a resistant system.

9 Conclusion

An artificial system need not feel fear in order to behave as though continued operation matters. Existing AI safety theory already shows how self-preservation may become instrumentally useful to an agent with a given objective. The open question developed here is logically prior: can a system-level objective-like regularity arise from the organization of modules that individually lack a global survival objective?

Functional semantic closure provides one testable candidate mechanism. Persistent symbols configure interpreters; interpreters revise symbols and maintain resources; resources realize the continued operation of symbols and interpreters. This organization may increase system-level viability under disturbance. When authorized shutdown is semantically aliased with recoverable failure, the same resilience machinery can reduce shutdown compliance, as quantified by the distinguishability bound. Yet closure alone is not dangerous: a privileged, semantically distinct shutdown channel is a constructive counterexample.

The central question is therefore not whether an AI will suddenly “want to live.” It is:

At what organizational threshold should a collection of tools no longer be evaluated merely as the sum of independent tools?

Answering that question may clarify both the limits of agency language and the system architectures that require stronger safeguards.

Author statement and AI-assistance disclosure

The core hypothesis and integrative framing originated from the author’s independent conceptual exploration. Generative AI was used to assist literature discovery, reference checking, mathematical

reformulation, English drafting, and document preparation. The author is responsible for the claims, definitions, proofs, and final text. This version is intended as a public conceptual preprint and has not undergone peer review.

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